

Soumitri Chattopadhyay

UC San Diego, CA, USA

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[Google Scholar](#) | [GitHub](#) | [LinkedIn](#)

RESEARCH INTERESTS

Methods: Foundational Models, Multi-modal learning (Vision+X), In-Context learning, Visual Agentic AI

Applications: Medical Image Computing, Longitudinal Analysis, Clinical Workflows; Image Restoration

EDUCATION

University of California, San Diego

Ph.D. in Computer Science and Engineering

California, USA

Sept. 2025 – May 2028 (Expected)

University of North Carolina at Chapel Hill

M.S. in Computer Science (Ph.D. Transferred to UCSD)

North Carolina, USA

Aug. 2023 – May 2025

Jadavpur University

B.E. in Information Technology | CGPA (out of 10): 9.01

Kolkata, India

Aug. 2019 – June 2023

RESEARCH EXPERIENCES

UC San Diego

Graduate Student Researcher

California, USA

Sept. 2025 – Present

- Part of the [UCSD Biomedical Image Analysis Group](#), advised by [Prof. Marc Niethammer](#).
- Currently working on in-context promptable foundational models for multimodal 2D/3D medical segmentation.
- Developed cross-modal visual prompting for medical segmentation with foundational models (**MICCAI '26**).
- Analyzed robustness of segmentation foundational models against imprecise visual prompts (**ISBI '26 Oral**).
- Developed an end-to-end framework to process, segment and extract biomarkers from 2D and 3D raw clinical imaging data for early diagnosis of Lupus (as a part of an **NIH grant**).

UNC Chapel Hill

Graduate Research Assistant

North Carolina, USA

Aug. 2023 – Sept. 2025

- Part of the erstwhile [UNC-BIAG](#), advised by [Prof. Marc Niethammer](#). Transferred to UCSD upon advisor's move.
- Conducted an extensive study on zero-shot domain generalizability of medical segmentation foundational models.
- Proposed a few-shot latent diffusion model-aided fine-grained image recognition (**BMVC '24 Oral**).

SRI International

Research Intern

Menlo Park, CA

May 2024 – Dec. 2024

- Interned in the [NuSCI Research Group](#), supervised by [Dr. Anirban Roy](#)
- Worked on **brain decoding**; developed an end-to-end framework for visual decoding from fMRI scans with controllable latent diffusion models and various training recipes that outperformed prior SOTA works.

École de technologie supérieure (ÉTS), Montreal

MITACS Globalink Research Intern

Montreal, Canada

June 2022 – Aug. 2022

- Advisor: [Prof. Jose Dolz](#)
- Explored handling of 3D medical volumes (CT and MRI), including pre-processing, sampling, and visualization.
- Implemented a knowledge distillation-aided baseline for unpaired multi-modal medical image segmentation.

CVPR Unit, Indian Statistical Institute

Research Intern

Kolkata, India

Nov. 2021 – May 2023

- Supervisors: [Prof. Umapada Pal](#) and [Prof. Saumik Bhattacharya](#)
- Investigated the role of temperature in Self-supervised Contrastive loss and proposed a dynamic temperature scaling to improve its performance (**IEEE TAI '24**).
- Developed Self-Supervised Learning frameworks for Offline Signature Verification (**ICIP '22 Oral, ICPR '22**).

3. **S. Chattopadhyay**, B. Demir, Y. Jin, M. Niethammer, [MedSegQC: A Data-driven Approach for Quality Assessment of 3D Medical Image Segmentation](#), *Under Review*.
2. D. Mandal, **S. Chattopadhyay**, Y. Wang, M. Niethammer, P. Chakravarthula, [UNSCAR: Universal, Scalable, Controllable, and Adaptable Image Restoration](#), *Under Review*.
1. **S. Chattopadhyay**, Y. Wang, D. Mandal, B. Demir, M. Niethammer, P. Chakravarthula, [Diffractive Optical Neural Networks for Ultra-Light-Weight Medical Image Segmentation](#), *Under Review*.

SELECTED PUBLICATIONS

13. **S. Chattopadhyay**, B. Demir, M. Niethammer, [Uncertainty-Aware Spatio-Semantic Contextual Prompts for Multimodal Medical Segmentation](#), **MICCAI 2026** (*Early Accepted, top-9%*)
12. B. Demir, **S. Chattopadhyay**, H. Greer, B. Chen, M. Niethammer, [How Useful Are Vision Foundation Model Features for Out-of-the-Box Disease Progression Prediction?](#), **CVPR 2026 – CV4Clinic Workshop, Best Paper Award**
11. A. Das, S. Biswas, **S. Chattopadhyay**, A. Lodh, et al., [NOTESBANK: Benchmarking Neural Transcription and Search for Scientific Notes Understanding](#), **ECML-PKDD 2026**
10. **S. Chattopadhyay**, B. Demir, M. Niethammer, [On the Robustness of Foundational 3D Medical Image Segmentation Models Against Imprecise Visual Prompts](#), **ISBI 2026, Oral**
9. D. Mandal, **S. Chattopadhyay**, G. Tong, P. Chakravarthula, [UNICORN: Latent Diffusion-based Unified Controllable Image Restoration Network across Multiple Degradations](#), **WACV 2026, Oral**
8. **S. Chattopadhyay**, S. Biswas, E. Vivoli, J. Lladós, [Towards Generative Class Prompt Learning for Fine-grained Visual Recognition](#), **BMVC 2024, Oral**
7. A. Sain, A.K. Bhunia, S. Koley, P.N. Chowdhury, **S. Chattopadhyay**, T. Xiang, Y.Z. Song, [Exploiting Unlabelled Photos for Stronger Fine-Grained SBIR](#), **CVPR 2023**
6. H. Thakur and **S. Chattopadhyay**, [Active Learning for Fine-Grained Sketch-Based Image Retrieval](#), **BMVC 2023**
5. M. Bhattacharyya, **S. Chattopadhyay**, S. Nag, [DeCAtt: Efficient Vision Transformers with Decorrelated Attention Heads](#), **CVPR 2023 – ECV Workshop, Oral**
4. **S. Chattopadhyay**, S. Ganguly, S. Chaudhury, S. Nag, S. Chattopadhyay, [Exploring Self-Supervised Representation Learning For Low-Resource Medical Image Analysis](#), **ICIP 2023**
3. H. Basak, **S. Chattopadhyay**, R. Kundu, S. Nag, R. Mallipeddi, [IDEAL: Improved DENSE locAL Contrastive Learning for Semi-Supervised Medical Image Segmentation](#), **ICASSP 2023**
2. S. Manna, **S. Chattopadhyay**, S. Bhattacharya, U. Pal, [SWIS: Self-Supervised Representation Learning For Writer Independent Offline Signature Verification](#), **ICIP 2022, Oral**
1. **S. Chattopadhyay**, S. Manna, S. Bhattacharya, U. Pal, [SURDS: Self-Supervised Attention-guided Reconstruction and Dual Triplet Loss for Offline Signature Verification](#), **ICPR 2022**

ACADEMIC SERVICES

Peer-reviewing (Regular, since 2022)

- **Conferences:** CVPR, MICCAI, WACV, ISBI, ICCV, ACM MM, ICPR, BMVC, ACCV, NeurIPS
- **Journals:** Medical Image Analysis - Elsevier ; Engineering Applications of Artificial Intelligence - Elsevier ; Computers in Biology and Medicine - Elsevier ; IEEE Access